

Predictive Maintenance for Hospital MRI Machine Uptime Using Sensor Data

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Problem

Predicting maintenance needs for MRI scanner hardware to reduce downtime.

Method

Time-series anomaly detection applied to sensor logs from MRI machines.

Dataset

Proprietary hospital equipment sensor logs.

Evaluation

Precision and recall of failure prediction.

Results

The anomaly detector predicted 78% of failures at least 48 hours in advance.

Limitations

Mentions 'MRI' but concerns equipment reliability, not diagnostic image analysis - a keyword-collision distractor for topic filtering.